

Working with MongoDB

1. Perform CRUD operations for a simple application to maintain database named 'company' and a collection named 'employees' inside it.
 - a. Insert multiple documents into the 'Employees' collection with sample employee data. Create your own schema for 'Employee'.
 - b. Retrieve all employees who work in the 'Engineering' department from MongoDB.
 - c. Update 'Bob's age in the 'employees' collection.
 - d. Delete the employee 'Charlie' from the 'employees' collection.
 - e. **Aggregate methods and Expression operators:**
Consider an ecommerce database with an orders collection.
Fields: customer_id, items (array of objects with prod_id and price), order_date, and status.
 - i. Calculate the total revenue generated from all "delivered" orders.
 - ii. Find the top 3 customers who have spent the most expense in the year 2025.
 - iii. Use \$unwind to list the total quantity sold for each unique prod_id across all orders.
 - iv. Group orders by month and calculate the average order value for each month.
 - f. **Indexing:** A query needs to be executed which searches for student information by branch and then sorts by CGPA (descending), is running slow. Write a command to create a compound index to optimize the query.
 - g. Write queries to find documents using Comparison operators and Array operators.
2. Write a Python script to connect to MongoDB and retrieve all employees' data from MongoDB and convert it into a Pandas DataFrame.
 - a. Retrieve employee data, encode the 'department' column using LabelEncoder, and prepare it for model training.
 - b. Train a Linear Regression model to predict an employee's age based on their department encoding.

Learning Resources:

- a. <https://www.mongodb.com/docs/manual/crud/>
- b. <https://mongodb.github.io/node-mongodb-native/2.2/quick-start/quick-start/>
- c. <https://www.w3schools.com/mongodb/>

Moodle Submission Logistics:

- You need to have the following files in a zipped folder with the naming format: "Mongodb_<<roll_no>>"
 - (a) Your code files
 - (b) Your datasets
 - (c) Your output files arranged in proper order
 - (d) any reference materials you have used

---THE END---